Mandatory Assignment 2

Astrid Haarup Møller, Esmeralda Ophelia Larsson & Nanna Munch Kjøller

April 11, 2026

Exercise 1

1a: Variable selection and sample restriction

First, we remove observations before year 2000, then we select ten stock-level characteristics based on the variable-importance results in Gu et al. (2020), Figure 5. The selected variables cover price trends: `mom1m`, `mom12m`, size and liquidity: `mvel1`, `turn`, risk and volatility: `beta`, `retvol`, and valuation/fundamentals: `bm`, `ep`, `roaq`, `invest`. For the macroeconomic state variables from Goyal and Welch (2008) library, we follow Gu et al. (2020) Table 4 and include aggregate book-to-market (`bm`), the Treasury bill rate (`tbl`), the default yield spread (`dfy`), and the term spread (`tms`). These variables capture aggregate valuation, interest rate conditions, credit risk, and the slope of the yield curve.

1b: Summary Statistics and Missing Values

Table 1 reports summary statistics for the selected predictors. Stock characteristics are approximately centered around zero and bounded close to $[-1, 1]$, while macro variables vary at the monthly aggregate level.

Table 1: Summary statistics for selected predictors

	mean	std	25%	50%	75%
characteristic_mom1m	-0.004	0.604	-0.550	-0.009	0.544
characteristic_mom12m	-0.007	0.590	-0.526	-0.026	0.519
characteristic_mvel1	0.026	0.588	-0.496	0.044	0.544
characteristic_turn	0.036	0.576	-0.453	0.074	0.532
characteristic_beta	0.076	0.548	-0.370	0.102	0.540
characteristic_retvol	0.123	0.528	-0.306	0.146	0.576
characteristic_bm	-0.065	0.531	-0.512	-0.067	0.373
characteristic_ep	-0.011	0.558	-0.486	-0.004	0.454
characteristic_roaq	-0.014	0.560	-0.477	-0.023	0.457
characteristic_invest	-0.006	0.533	-0.427	-0.000	0.413
macro_bm	0.276	0.072	0.239	0.284	0.328
macro_tbl	0.018	0.019	0.001	0.011	0.028
macro_dfy	0.010	0.004	0.008	0.009	0.012

Table 1: Summary statistics for selected predictors

	mean	std	25%	50%	75%
macro_tms	0.022	0.014	0.009	0.023	0.035

Missing stock characteristics are imputed using the monthly cross-sectional median. This preserves the number of stock-month observations while using only contemporaneous information from the same month. The median is preferred to the mean because it is less sensitive to extreme firm-level observations, which are common in financial data.

Macroeconomic variables are common to all firms in a given month and vary only over time. Missing macro values are therefore filled forward and backward to preserve the time series of aggregate state variables. Finally, observations with missing excess returns or lagged market capitalization are dropped because `ret_excess` is the prediction target and `mktcap_lag` is required to construct value-weighted portfolio returns.

We recode `sic2` into the 10 major SIC divisions to retain broad industry information while keeping the feature space manageable. This reduces the number of industry dummy variables and lowers the risk that sparse industry categories create noisy predictors in the machine learning models.

Overall, the sample remains large after cleaning and is suitable for machine learning. The missing-value treatment balances data retention with robustness, while the industry recoding keeps the dimensionality of the feature matrix manageable.

1c: Feature matrix construction

We construct the feature matrix using the selected stock characteristics and their interactions with macroeconomic state variables. The feature matrix contains 59 predictors: 10 stock characteristics, 40 characteristic-macro interactions, and 9 industry dummies. The interaction terms allow firm-level return predictability to vary with macroeconomic conditions, consistent with Gu et al. (2020). One SIC division is omitted as the reference category when constructing industry dummies.

Exercise 2

Exercise 2a: Model descriptions

Elastic Net is a penalized linear regression model that combines the variable selection property of the Lasso with the coefficient shrinkage of Ridge regression. It is useful in our setting because the feature matrix contains many correlated predictors, including characteristic-macro interactions. The model minimizes prediction error while penalizing large coefficients, which helps reduce overfitting.

The key hyperparameters are the overall penalty strength (`alpha`) and the mixing parameter (`l1_ratio`). A higher `alpha` implies stronger regularization, while `l1_ratio` controls the balance between Lasso and Ridge. When `l1_ratio = 1`, the model corresponds to the Lasso; when `l1_ratio = 0`, it corresponds to Ridge regression.

Random Forest is a nonlinear tree-based ensemble model. It builds many regression trees on different bootstrap samples of the data and averages their predictions. Each tree recursively splits the feature space into groups with similar realized excess returns. Averaging across trees reduces the instability of individual decision trees and helps improve out-of-sample performance.

The key hyperparameters are the number of trees (`n_estimators`), the number of predictors considered at each split (`max_features`), the minimum number of observations required in a leaf (`min_samples_leaf`), and tree depth (`max_depth`). Compared with Elastic Net, Random Forest is more flexible because it can capture nonlinearities and interactions without imposing a linear functional form.

Together, the two models provide a comparison between a regularized linear benchmark and a flexible nonlinear machine learning model.

Exercise 2b: Time-respecting data split

	Sample	Start date	End date	Months	Observations
0	Train	2000-01-01	2013-02-01	158	748902
1	Validation	2013-03-01	2017-07-01	53	194460
2	Test	2017-08-01	2021-12-01	53	168243

The dataset is split chronologically to avoid look-ahead bias. Training observations precede validation observations, which in turn precede test observations. The most recent 20% of months are reserved for the test set and are only used for final out-of-sample evaluation.

The remaining 80% of months is split into training and validation samples, yielding an approximate 60/20/20 split. The training set is used to estimate model parameters, while the validation set is used for hyperparameter tuning. The split is performed by month rather than by individual rows to preserve the panel structure.

Continuous predictors are standardized using the training sample only. The same transformation is then applied to the validation and test sets, which avoids data leakage from future observations.

Exercise 2c: Hyperparameter tuning

	alpha	l1_ratio	train_rmse	val_rmse
6	0.100	0.2	0.199689	0.157523
7	0.100	0.5	0.199689	0.157523
5	0.010	0.8	0.199689	0.157523
8	0.100	0.8	0.199689	0.157523
4	0.010	0.5	0.199682	0.157524
3	0.010	0.2	0.199484	0.157563
0	0.001	0.2	0.199279	0.157587
2	0.001	0.8	0.199378	0.157640
1	0.001	0.5	0.199346	0.157652

alpha	l1_ratio	train_rmse	val_rmse

	n_estimators	max_features	min_samples_leaf	max_depth	train_rmse	val_rmse
2	50	sqrt	100	8	0.196722	0.157406
1	50	sqrt	100	5	0.198013	0.157424
0	50	sqrt	100	3	0.198846	0.157455

	Model	Train RMSE	Validation RMSE
0	Elastic Net	0.199689	0.157523
1	Random Forest	0.196722	0.157406

	Model	alpha	l1_ratio	n_estimators	max_features	min_samples_leaf	max_depth
0	Elastic Net	0.1	0.2	NaN	NaN	NaN	NaN
1	Random Forest	NaN	NaN	50.0	sqrt	100.0	8.0

Hyperparameters are tuned using grid search. Each candidate specification is fitted on the training sample and evaluated on the validation sample using RMSE, consistent with the mean squared prediction error objective in Gu et al. (2020). For Elastic Net, the grid varies the overall penalty strength (`alpha`) and the balance between Lasso and Ridge regularization (`l1_ratio`). For Random Forest, the grid varies the number of predictors considered at each split (`max_features`), the minimum leaf size (`min_samples_leaf`), and the maximum tree depth (`max_depth`).

The final specification for each model is selected as the one with the lowest validation RMSE. Comparing training and validation RMSE illustrates the bias-variance trade-off: more flexible models may fit the training data better, but this does not necessarily translate into better validation performance.

Exercise 2d: Save models to disk

To load the models for peer review, run:

```
final_en = joblib.load("model_elastic_net.joblib")
final_rf = joblib.load("model_random_forest.joblib")
```

The final fitted models are saved to disk using `joblib` to ensure reproducibility and facilitate peer review. Saving the trained models allows others to load the exact same fitted objects without re-running the full hyperparameter tuning procedure.

Exercise 3

Exercise 3a-c: Predictions and long-short portfolios

Using the fitted Elastic Net and Random Forest models from Exercise 2, we generate predicted excess returns for each stock-month observation in the validation and test samples. The test set is only used for final out-of-sample evaluation and is not used for model fitting or hyperparameter selection.

For each month, stocks are sorted into decile portfolios based on predicted excess returns. Decile 0 contains the lowest predicted returns and decile 9 contains the highest predicted returns. Realized excess returns within each decile are computed as value-weighted averages using `mktcap_lag`.

The long-short strategy buys the top predicted-return decile and shorts the bottom predicted-return decile. The monthly long-short return is therefore the value-weighted return of decile 9 minus the value-weighted return of decile 0.

Exercise 3d: Performance metrics

The market benchmark is constructed as the value-weighted average excess return across all stocks in the relevant sample. This makes the benchmark comparable to the long-short portfolio, since both returns are measured in excess of the risk-free rate. CAPM alpha is estimated by regressing the long-short portfolio excess return on the value-weighted market excess return.

Table 2 reports annualized mean excess returns, Sharpe ratios, CAPM alphas, and alpha t-statistics for the long-short portfolios and the market benchmark. Elastic Net performs consistently across validation and test, with positive annualized returns and Sharpe ratios in both periods. Random Forest performs reasonably in validation but delivers negative test-period performance, suggesting that its predictive signal is less stable out-of-sample. The alpha t-statistics are below conventional significance levels, so the evidence for statistically significant abnormal returns is weak.

Table 2: Annualized long-short portfolio performance and market benchmark

	Model	Sample	ann_mean	sharpe	alpha	alpha_t
0	Elastic Net	Validation	0.0410	0.6976	0.0405	1.3369
1	Random Forest	Validation	0.0364	0.3143	0.0702	1.2036
2	Market	Validation	0.1361	1.3293	NaN	NaN
3	Elastic Net	Test	0.0641	0.7228	0.0449	1.0329
4	Random Forest	Test	-0.0321	-0.1777	-0.0369	-0.4066
5	Market	Test	0.1748	1.0233	NaN	NaN

Exercise 3e: Portfolio turnover

Table 3 reports average monthly turnover.

Table 3: Average monthly portfolio turnover

	Model	Validation	Test
0	Elastic Net	0.007	0.008
1	Random Forest	0.542	0.501

Exercise 3f: Model comparison

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For the selected hyperparameter specifications, Elastic Net is the preferred model from an investor perspective. In the test period, Elastic Net generates an annualized mean return of 6.41%, a Sharpe ratio of 0.72, and a CAPM alpha of 4.49%. Random Forest performs poorly out-of-sample, with an annualized mean return of -3.21%, a Sharpe ratio of -0.18, and a CAPM alpha of -3.69%.

Table 3 shows that Elastic Net also has substantially lower turnover. In the test period, Elastic Net turnover is 0.008, compared with 0.501 for Random Forest. This matters because high turnover increases transaction costs and can erode gross portfolio returns. Overall, Elastic Net provides the better trade-off between risk-adjusted performance, alpha, and implementability.

Exercise 4

Exercise 4a: Variable importance

	model	feature	sharpe_drop
0	Elastic Net	characteristic_mom1m	0.0000
1	Elastic Net	characteristic_mom12m	0.0000
2	Elastic Net	characteristic_mvel1	0.0000
3	Random Forest	characteristic_ep_x_macro_bm	0.4633
4	Random Forest	characteristic_retvol_x_macro_tbl	0.2944
5	Random Forest	characteristic_mvel1_x_macro_dfy	0.2885

Variable importance is measured as the drop in annualized test Sharpe ratio when one predictor is set to a reference value and the long-short portfolio is rebuilt. For standardized continuous predictors, the reference value is zero, corresponding to the training-sample mean after standardization. For industry dummies, the reference value is set to the test-sample mean.

This portfolio-based importance measure evaluates whether a predictor matters for the final investment strategy rather than only for return prediction accuracy. A larger Sharpe-ratio drop indicates that removing the information in that predictor reduces the economic value of the model.

Exercise 4b: Ranking comparison

	feature	rank_en	rank_rf	rank_gap
5	characteristic_bm	1.0	59.0	58.0
33	characteristic_mvel1_x_macro_tbl	1.0	58.0	57.0
37	characteristic_retvol_x_macro_dfy	1.0	57.0	56.0
22	characteristic_mom12m_x_macro_dfy	1.0	56.0	55.0
36	characteristic_retvol_x_macro_bm	1.0	55.0	54.0

The rankings differ across the two models because Elastic Net and Random Forest use the predictors in different ways. Elastic Net is a regularized linear model, so predictor importance is tied to linear effects after shrinkage. Random Forest is more flexible and can assign importance to predictors that matter through nonlinear patterns or interactions. Differences in the rankings therefore reflect differences in model structure, not only differences in predictive strength.

Exercise 4c: Findings to financial theory

The variable-importance results can be interpreted in relation to financial theory. If momentum-related variables rank highly, this is consistent with return continuation or reversal effects. If valuation variables such as book-to-market or earnings-to-price are important, the results point toward value-based predictability. Risk-related variables such as beta or return volatility can be interpreted as compensation for systematic or idiosyncratic risk, while industry dummies suggest that expected returns vary across sectors.

Because Random Forest can capture nonlinearities and interactions, its most important variables may reflect state-dependent effects rather than simple standalone return premia. Overall, the variable-importance exercise links the machine-learning predictions back to economically interpretable return predictors.

References

Goyal, A. and Welch, I. (2008). A comprehensive look at the empirical performance of equity premium prediction. *The Review of Financial Studies*, 21(4):1455–1508.

Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273.